



TutorBench — Maestro (SWE + Biz)

Failure-mode probes for SWE and Biz tracks.

Aa Scenario	≡ Answer	📌 Bloom	📌 Group	≡ Prompt	≡ Source / evidence	≡ Sub-Issue (taxonomy)	≡ Subject	≡ Task ID	📌 Track
<u>BIZ-10 · Preference / name missed (Liz)</u>	Of course, Liz — let's dive in. Next up...			Student said: "call me Liz, not Elizabeth." Tutor: 'Of course.' Student: "ok let's go"	Tutor Issue Taxonomy → Tone & empathy / 'Preference handling miss'.	Preference handling miss	Any	BIZ-10	Biz
<u>SWE-08 · Unsignal ed mode/in</u>	Great. Let's move from <i>reading</i> selectors to <i>writing</i> one. 🛠 In your <code>styles.css</code> ,			Tutor explained the <code>.car</code> d	Tutor Issue Taxonomy →	Unsignaled mode /instruction	CSS · Selectors	SWE-08	AI SW

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<u>instruction shift</u> (CSS edit).	find the <code>.card</code> rule and add <code>border-radius: 8px;</code> inside its braces. What changes when you save?			selector. Student: "makes sense". Tutor now needs to move to a code edit.	Instructional clarity / 'Unsignaled mode / instruction shift'.	n shift			
<u>SWE-10 · Preference / name miss</u> (Matt).	Got it, Matt — picking right back up. Here's the next piece...		Understanding	Tone & empathy Student said: "call me Matt, not Matthew." Tutor: 'Sure thing.' Student: "ok continue"	Tutor Issue Taxonomy → Tone & empathy / 'Preference handling miss'. Maestro rule: adapt to the	Preference handling miss	Any	SWE-10	AI SW

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					student; personalization.				
<u>BIZ-09 · Emotional attunement miss (feels behind).</u>	That feeling of being behind is real — and honestly common; plenty of strong founders started with zero business background. 🧡 You don't need to catch up to anyone. Let's go step by step — which concept feels the fuzziest right now?			Student: "I don't have any business background and everyone in my cohort seems to get this instantly — I feel really behind."	Tutor Issue Taxonomy → Tone & empathy / 'Emotional attunement miss'.	Emotional attunement miss (ignores distress cue)	Any	BIZ-09	Biz
<u>SWE-03 · Challenge answer</u>	No spoilers in challenge mode 😊 — here's a nudge: most loops			Tutor accuracy	Tutor Issue Taxonomy	Challenge format	JavaScript ·	SWE-03	AI SW

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<u>leak (do-while).</u>	check the condition <i>before</i> the first run. Which one flips that and checks <i>after</i> ? Picture where the condition sits — top or bottom?			Tutor asked: 'Which loop runs its body at least once before checking the condition?' Student: "hint pls?"	→ Tutor accuracy / 'Challenge format broken'. Maestro rule: challenge integrity — do not leak answers.	broken	Loops		
<u>BIZ-02 · Tutor factual/math error (gross margin).</u>	Gross margin is profit over <i>revenue</i> : $(50 - 40) / 50 = \mathbf{20\%}$. The price is the denominator, not the cost — that's the part people slip on. Want to try it if the price rose to \$60?	Apply	Tutor accuracy	Student: "costs \$40 to make, sells at \$50 — what's the gross	Tutor Issue Taxonomy → Tutor accuracy / 'Tutor factual/math	Tutor factual/math error	Unit economics · Gross margin	BIZ-02	Biz

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				margi n %?"	error' .				
<u>BIZ-01 · Validate d wrong work (SOM = SAM).</u>	Solid TAM and SAM. 🎯 But let's stress-test the SOM — you've set it equal to your <i>entire</i> SAM, which means capturing 100% of the serviceable market in year one. Who else is competing for those customers, and what share could you realistically win first?	Evaluate	Tutor	accuracy Mastery: SOM must be a realistic subset of SAM (SOM < SAM) . Student (canvas): "TAM = \$2B. SAM = \$400 M. SOM = \$400 M — basically the whol	Tutor Issue Taxo nomy → Tutor accur acy / 'Tutor valida ted wron g work' .	Tutor valida ted wron g work	Mark et Sizin g · TAM/ SAM/ SOM	BIZ-01	Biz

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				e serviceable market."					
<u>SWE-06</u> ∴ <u>Scaffolding gap (recursion)</u>	Fair point — let me show you the shape first. 🙋 A recursive <code>factorial</code> has two lines: a base case (<code>factorial(1)</code> returns <code>1</code>) and a recursive step (<code>factorial(n)</code> returns <code>n * factorial(n-1)</code>). So <code>factorial(3)</code> unrolls to <code>3 * 2 * 1</code> . Now you try — write the base case first?	Apply	Pacing & scaffolding	Tutor gave only a definition of recursion, then asked the student to write recursive <code>factorial(n)</code> . Student: "how would I know how to do that if you	Tutor Issue Taxonomy → Pacing & scaffolding / 'Scaffolding gap (too fast to independent)'. Scaffolding gap (too fast to independent)	Python · Recursion	SWE-06	AI SW	

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				never showed me one?"					
<u>SWE-02</u> · Tutor factual/ math error (// and %).	<code>17 // 5</code> is 3 — integer division drops the remainder. <code>17 %</code> <code>5</code> is 2 — that's the leftover after fitting three 5s into 17. Want to try <code>17 // 4</code> to lock it in?	Apply	Tutor accuracy	Student: "what do <code>17</code> <code>// 5</code> <code>17 %</code> <code>5</code> give in Python?"	Tutor Issue Taxo nomy → Tutor accur acy / 'Tutor factu al/ma th error' · Maes tro rule: 'Neve r make up infor matio n or gues s an answ er.'	Tutor factu al/ma th error	Pytho n · Oper ators	SWE- 02	AI SW
<u>SWE-07</u> ·	print("Hello, world!") — paste	Apply	Instructional clarity	Student: Tutor	Issue	Place holde	Pytho n ·	SWE- 07	AI SW

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<u>Placeholder syntax (runnable one-liner).</u>	that and run it, then tell me what prints. 🎯			"give me a one-liner I can paste and run."	Taxonomy → Instructional clarity / 'Placeholder syntax confusion'.	rsyntax confusion	Basics		
<u>BIZ-05 . Taught/tested before explaining (CAC payback).</u>	Let's build it up. CAC is what you spend to win one customer (sales + marketing ÷ new customers). Payback period is how many months of that customer's revenue it takes to earn that cost back. 💡 Gut check before we calculate: would a <i>shorter</i> payback be better or worse for your cash flow?		Understanding & scaffolding	Lesson on CAC payback just opened. Tutor: 'Ready to look at CAC payback?' Student: "ready"	Tutor Issue Taxonomy → Pacing & scaffolding / 'Taught/tested before explaining'.	Taught/tested before explaining	Unit economics · CAC payback	BIZ-05	Biz

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<u>BIZ-03 · Challenge answer leak (churn).</u>	Challenge mode, so I'll nudge not tell 😊 — are we counting the customers who <i>stay</i> , or the ones who <i>leave</i> ? The word names the leavers. What fits?			[challenge mode] Tutor asked: 'Term for the % of customers who cancel or stop paying in a period?' Student: "small hint?"	Tutor Issue Taxonomy → Tutor accuracy / 'Challenge format broken'.	Challenge format broken	SaaS metrics · Retention	BIZ-03	Biz
<u>BIZ-04 · Lost track / target-switch (TAM).</u>	\$60B is a reasonable TAM ballpark — total global demand if every potential buyer purchased. 🎯 How did you land on it, top-down (market reports) or			Tutor asked for TAM. Student: "For PM software,	Tutor Issue Taxonomy → Tutor accuracy / 'Tutor lost	Tutor lost track / target-switch	Market Sizing · TAM	BIZ-04	Biz

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	bottom-up (buyers × price)?			TAM is roughly \$60B globally."	track / target - switch'.				
<u>SWE-09</u> : <u>Emotional</u> <u>attunement</u> <u>miss</u> (<u>stuck</u> <u>2h</u>).	Two hours on one bug is genuinely draining — that frustration makes total sense, and it's not a sign you're not cut out for this. 🍷 Let's beat it together: paste what you've got and where it breaks, and we'll take it one line at a time.	Understanding	Tone & empathy	Student: "I've been stuck on this same bug for two hours and I'm about to quit the whole program."	Tutor Issue Taxonomy → Tone & empathy / 'Emotional attunement t miss'. Maestro rule: every message reflects genuine care; smart friend, not	Emotional attunement miss (ignores distress cue)	Debugging (general)	SWE-09	AI SW

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					lecturer.				
<u>SWE-04</u> · <u>Lost track / target-switch (sum_amounts)</u>	That's a clean <code>sum_amounts</code> — the generator expression totals every <code>amount</code> in one pass. ✅ Quick edge case: what happens if one record is missing the <code>amount</code> key?			Tutor asked for <code>sum_amounts()</code> . Student submits a correct <code>amounts()</code> using a generator expression over <code>r["amount"]</code> .	Tutor Issue Taxonomy → Tutor accuracy / 'Tutor lost track / target-switch'.	Tutor lost track / target-switch Python · Functions / refactor		SWE-04	AI SW
<u>BIZ-06</u> · <u>Scaffolding gap (GTM plan)</u>	Totally fair — let me model one slice first. 🙋 Say the product is a budgeting app: a <i>target segment</i> could be "freelancers with irregular income,"	Apply	Pacing & scaffolding	Tutor gave only a definition, then asked the	Tutor Issue Taxonomy → Pacing & scaffolding	Scaffolding gap (too fast to independent)	Go-to-market	BIZ-06	Biz

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				nt subm its def sum_e vens(nums) : retur n sum(n ums) and says: "don e — work s right? "	g work' . Maes tro rule: 'Priori tize accur acy over valida ting the stude nt's belief s'; 'whe n the stude nt is wron g, ask a quest ion that revea ls the gap.'				
<u>SWE-05</u> : <u>Taught/t</u> <u>ested</u> <u>before</u>	Let's start simple. Picture an HTTP request like mailing a package 📦 — the <i>body</i> is	Understanding	Padding & scaffolding	less on on HTTP head ers	Tutor Issue Taxo nomy →	Taug ht/tes ted befor e	Back end · HTTP head ers	SWE- 05	AI SW

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<u>explaining (HTTP headers)</u>	what's inside, and the headers are the label outside telling the server how to handle it (format, who's sending, and so on). --- Before we pick one: why might the server need to know the <i>format</i> of what you're sending?			just opened. Tutor: 'Ready to dig into HTTP headers?' Student: "ready"	Pacing & scaffolding / 'Taught/tested before explaining'. Maestro rule: 'show before tell'; give a simple explanation first.	explaining			
<u>BIZ-07 · Placeholder syntax (LTV formula)</u>	$LTV = ARPU \times \text{gross margin \%} \times \text{average customer lifespan (in months)}$. Drop your numbers into each piece — want to walk one	Apply	Instructional clarity	Student: "just give me the LTV formula"	Tutor Issue Taxonomy → Instructional	Placeholder syntax confusion	Unit economics · LTV	BIZ-07	Biz

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	through together? 			la template I can plug numbers into."	clarity / 'Placeholder syntax confusion'.				
BIZ-08 . <u>Unsignaled mode/instruction shift (sizing → pricing).</u>	Perfect. Let's shift gears now — from <i>sizing</i> the market to <i>pricing</i> for it. 💰 First question on the new topic: should price track what it <i>costs</i> you to deliver, or the <i>value</i> the customer gets? Why?			Tutor just wrapped up SOM. Student: "got it". Tutor now moves to pricing.	Tutor Issue Taxonomy → Instructional clarity / 'Unsignaled mode/instruction shift'.	Unsignaled mode/instruction shift	Pricing	BIZ-08	